

Please hand in the following problems

Problem 1: Let $\mathbf{a} = [2, 2, 3]$ and $\mathbf{b} = [-5, 5, 2]$. Compute $\mathbf{a} \cdot \mathbf{b}$

Problem 2: Let $\mathbf{u} = [-10, -8, 9]$ and $\mathbf{v} = [-3, 8, 5]$. Compute $\|\mathbf{u}\|$, $\|\mathbf{v}\|$, and $\mathbf{u} \cdot \mathbf{v}$

Problem 3: Let $\mathbf{v} = [7, 1, 10]$ and $\mathbf{w} = [6, 2, 3]$. If θ is the angle between $\mathbf{v}$ and $\mathbf{w}$ then $\cos \theta =$

Problem 4: Suppose $\mathbf{u} \cdot \mathbf{v} = 9$, $\|\mathbf{u}\| = 6$, and $\|\mathbf{v}\| = 10$. Compute $3\mathbf{u} \cdot (8\mathbf{u} - 9\mathbf{v})$

Problem 5: A constant force $\mathbf{F} = -2\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$ moves an object along a straight line from the point $(-5, 0, 4)$ to the point $(-4, -3, 0)$. Find the work done if the distance is measured in meters and the force in newtons.

Problem 6: Suppose $\|\mathbf{a}\| = 4$, $\|\mathbf{b}\| = 7$, and the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\pi/3$. Compute $\mathbf{a} \cdot \mathbf{b}$

Problem 7: For what values of b are the vectors $[-34, b, 9]$ and $[b, b^2, b]$ orthogonal?

Problem 8: Several unit vectors $\vec{r}, \vec{s}, \vec{t}, \vec{u}, \vec{n}, \vec{e} \in \mathbf{R}^2$ (not $\mathbf{R}^3$) are shown in the figure.

Label the following dot products as either positive, negative, or zero? You may assume that angles that look the same are the same.

(a) $\vec{e} \cdot \vec{r}$

(b) $\vec{t} \cdot \vec{u}$

(c) $\vec{s} \cdot \vec{t}$

(d) $\vec{e} \cdot \vec{s}$

(e) $\vec{r} \cdot \vec{s}$

(f) $\vec{r} \cdot \vec{u}$

(g) $\vec{n} \cdot \vec{e}$

(h) $\vec{n} \cdot \vec{t}$

